

AWS- DHCP Option Set

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How DHCP Work?

- **Dynamic Host Configuration Protocol**
- **Job Role:** Assigns unique IP address to hosts.
- Can exists as a Client or a Server

Steps in DHCP

1. DHCP Discover – Host looks for the DHCP server
2. DHCP offer -- Server offers IP address
3. DHCP Request – Host request/ACK for the IP to be leased.
4. DHCPACK – Server Provides the IP address, Subnetmask, Default Gateway and Leased time

UDP Ports used : Client: 68 and Server: 67

AWS DHCP Options Set

- It's the Default DHCP service to assign PRIVATE ip's to the EC2 instance.
- Also assigns an private DNS names local to the VPC.
- Options Supported with DHCP set
 - Can configure our custom DNS server
 - Can configure custom NTP server
 - Configure Netbios-name-servers
 - And netbios-node-type

Things to Remember

- NO Extra Charges for this DHCP service
- Once the DHCP set is created, we CANNOT modify them.
- We can have VPC with no DHCP options.
- VPC can have multiple DHCP options.

CLI Command

- Create a set of DHCP options for your VPC –

```
aws ec2 create-dhcp-options --dhcp-configuration "Key=domain-name-servers,Values=10.2.5.1,10.2.5.2 Key=domain-name,Values=example.com Key=netbios-node-type,Values=2"
```

- Associate DHCP options set with your VPC

```
aws ec2 associate-dhcp-options --dhcp-options-id dopt-d9070ebb --vpc-id vpc-a01106c2
```

- Delete DHCP options set

```
aws ec2 delete-dhcp-options --dhcp-options-id dopt-d9070ebb
```

Troubleshooting Route 53

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/troubleshooting-route-53.html>